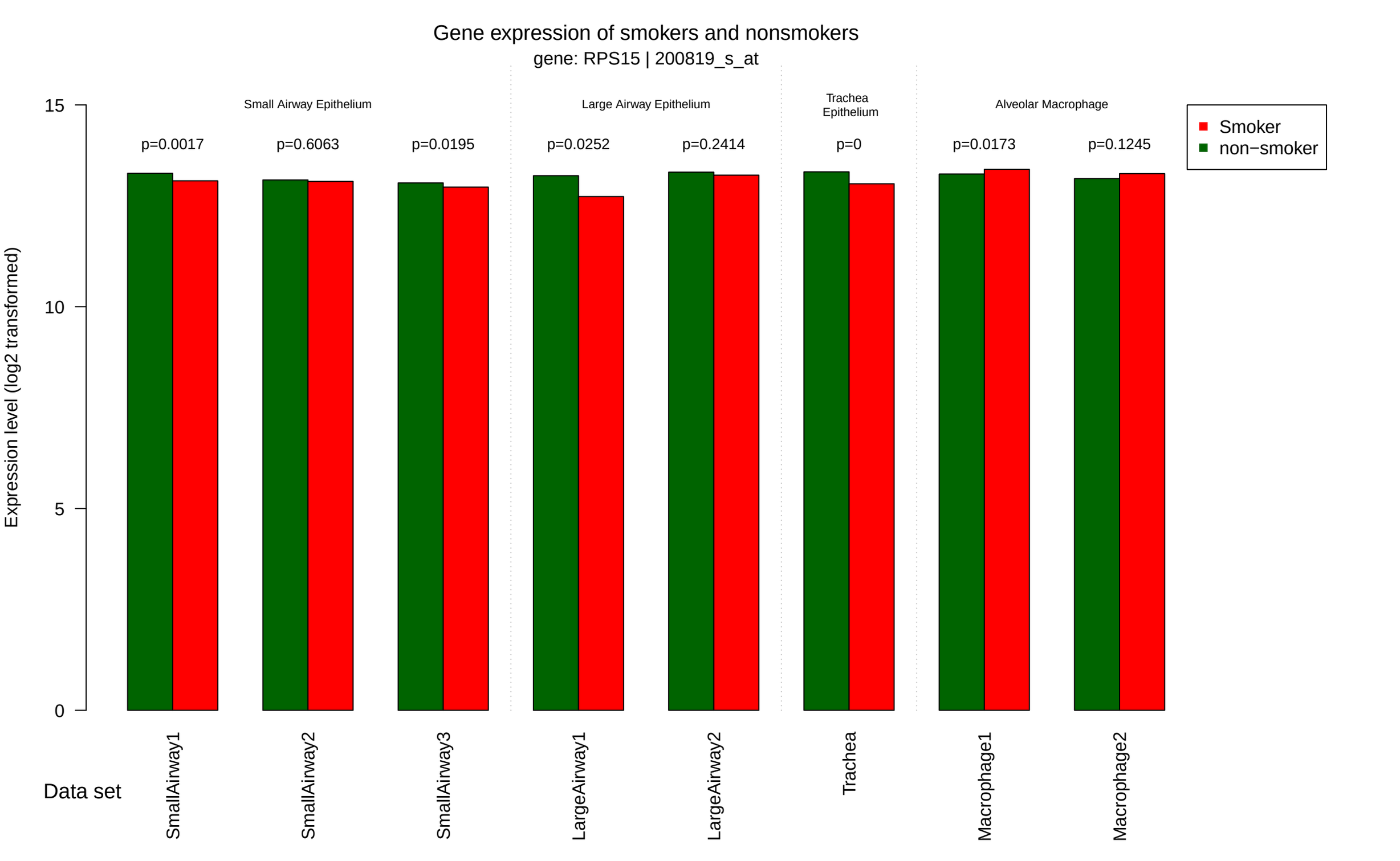
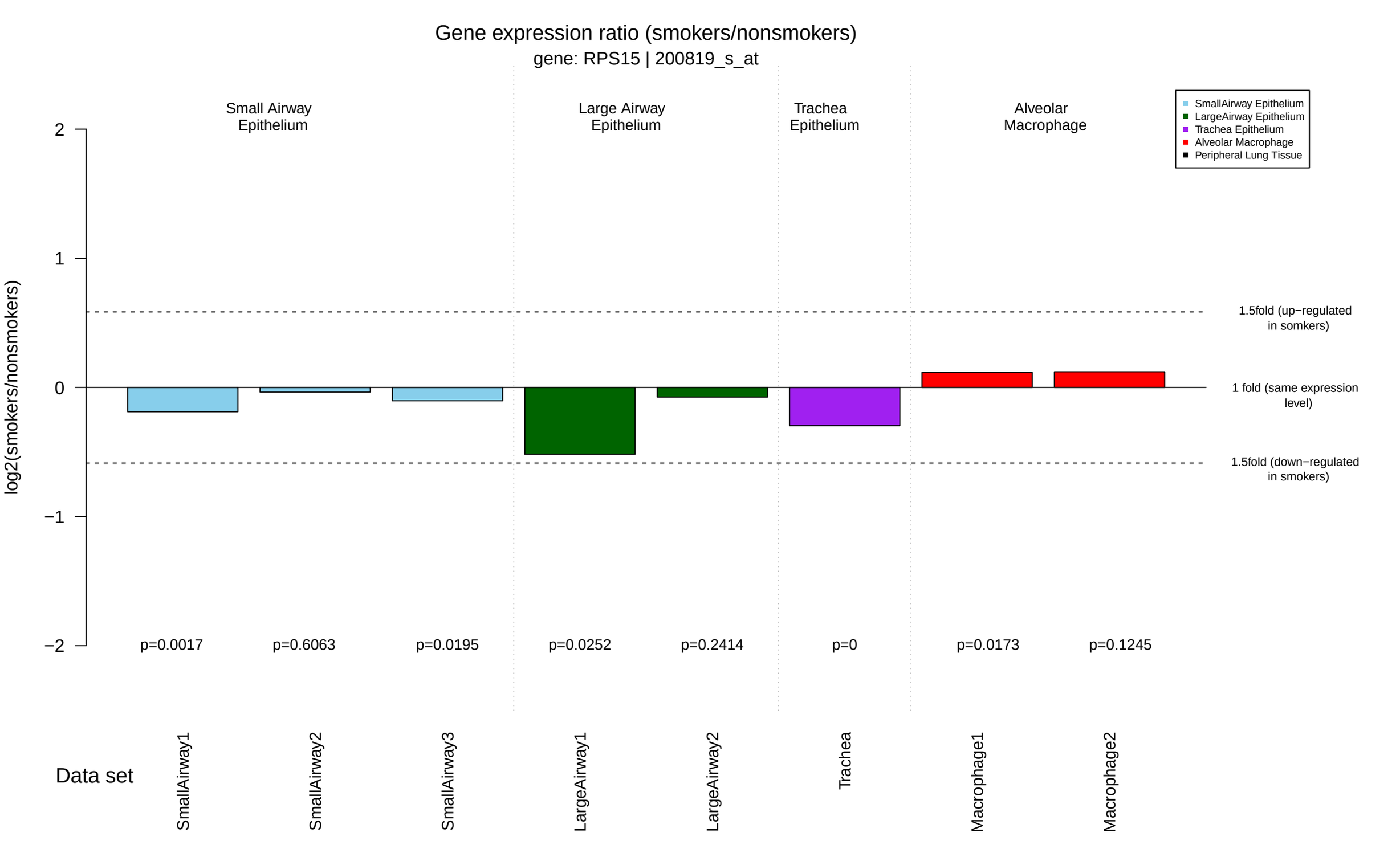
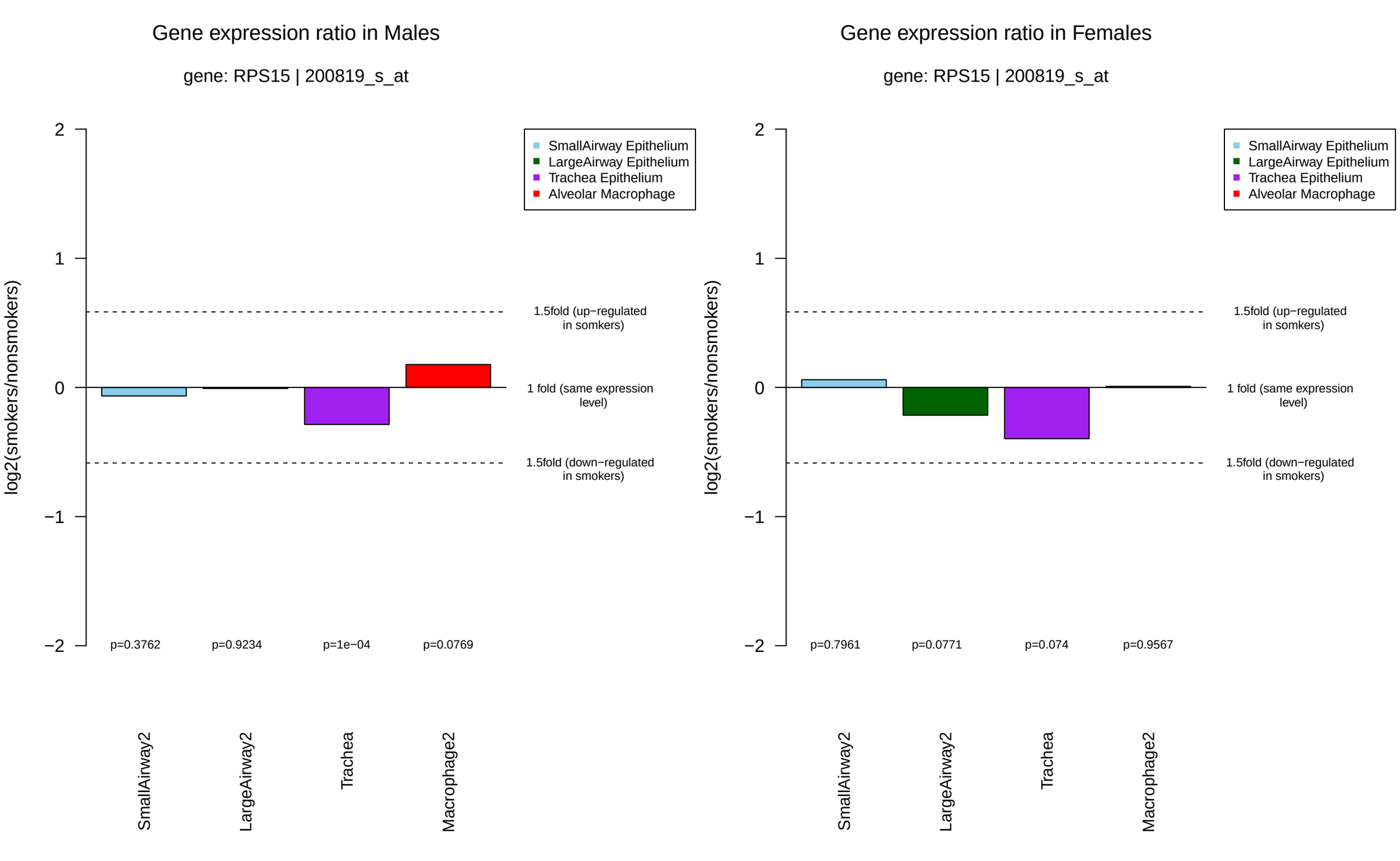
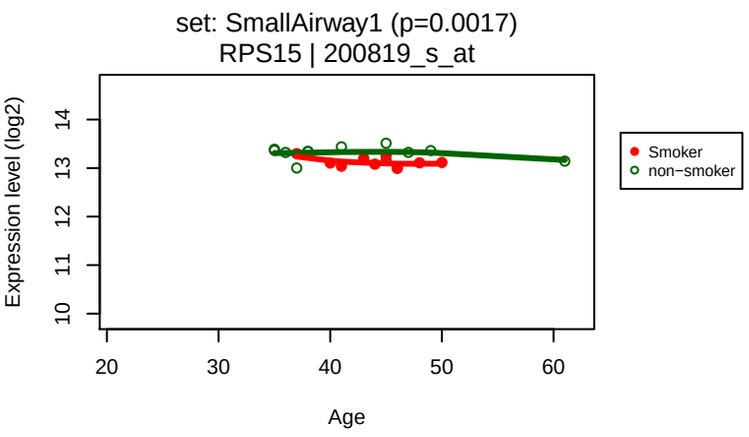
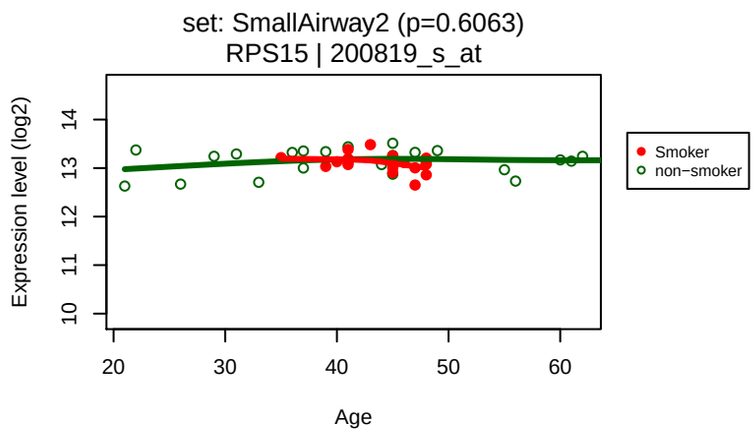
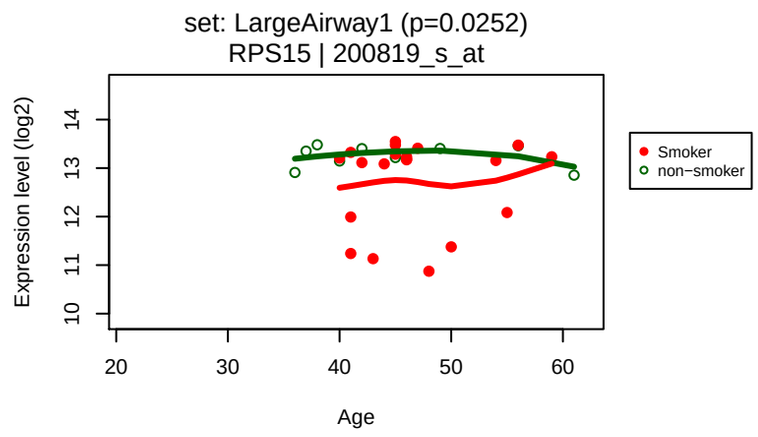

Gene expression of smokers and nonsmokers
gene: RPS15 | 200819_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0017
p=0.6063
p=0.0195
p=0.0252
p=0.2414
p=0
p=0.0173
p=0.1245
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

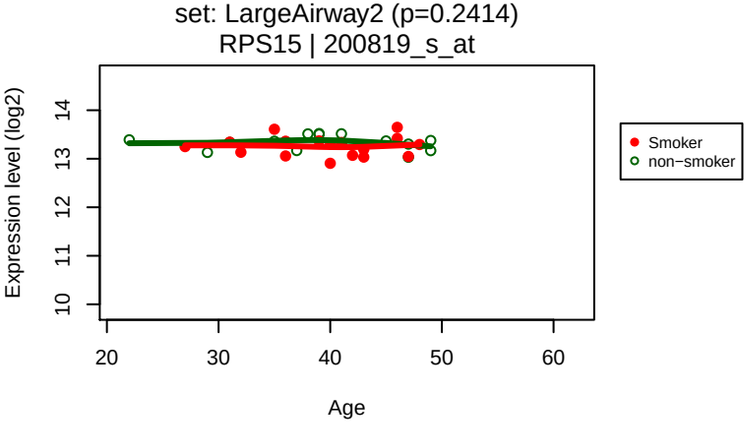

Gene expression ratio (smokers/nonsmokers)
gene: RPS15 | 200819_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0017
p=0.6063
p=0.0195
p=0.0252
p=0.2414
p=0
p=0.0173
p=0.1245
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

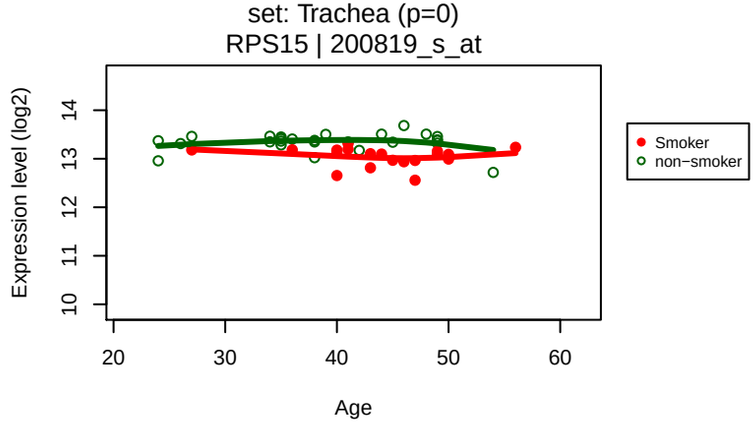

Gene expression ratio in Males
gene: RPS15 | 200819_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3762
p=0.9234
p=1e−04
p=0.0769
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS15 | 200819_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7961
p=0.0771
p=0.074
p=0.9567
SmallAirway2
LargeAirway2
Trachea
Macrophage2

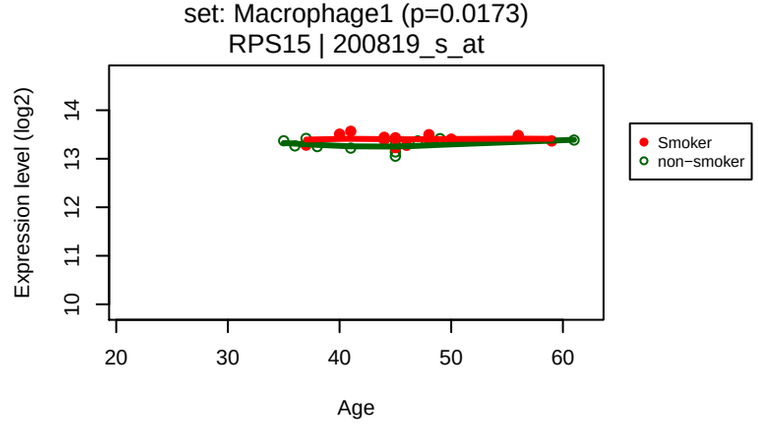

set: SmallAirway1 (p=0.0017)
RPS15 | 200819_s_at
Expression level (log2)
Age
Smoker
non−smoker

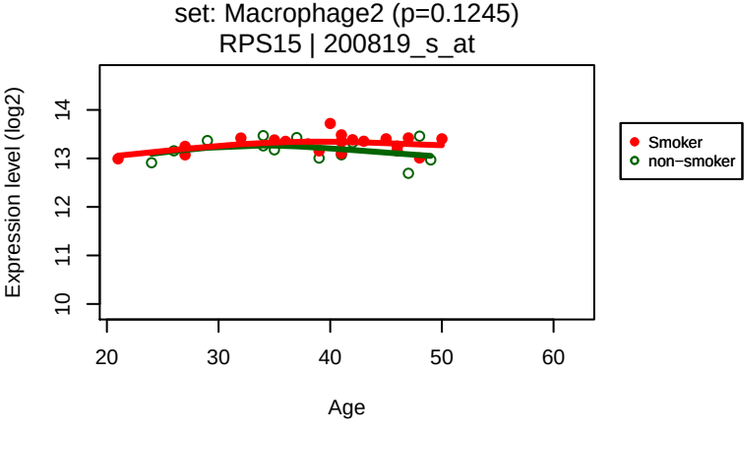

set: SmallAirway2 (p=0.6063)
RPS15 | 200819_s_at
Expression level (log2)
Age
Smoker
non−smoker

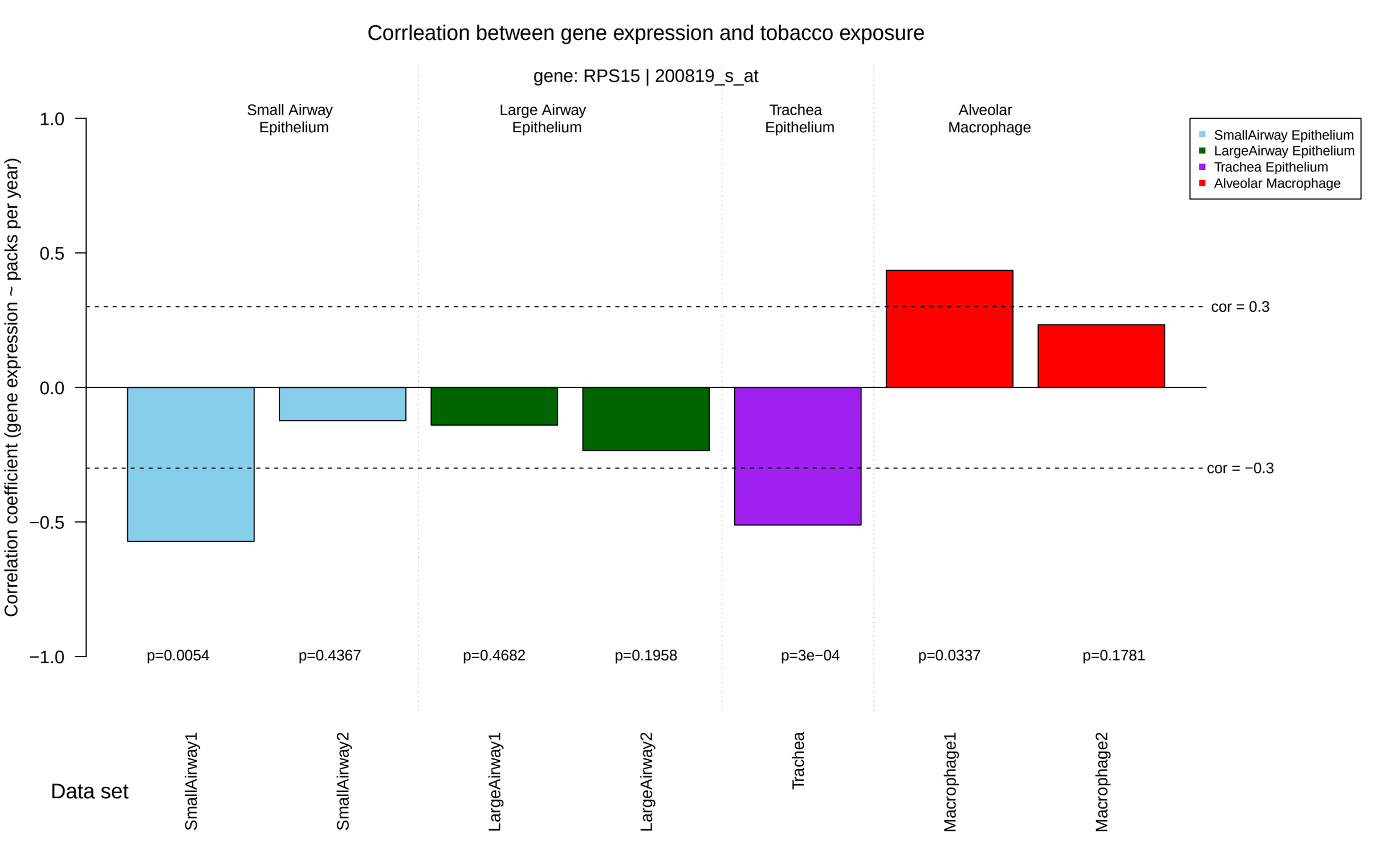

set: LargeAirway1 (p=0.0252)
RPS15 | 200819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2414)
RPS15 | 200819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
RPS15 | 200819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0173)
RPS15 | 200819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1245)
RPS15 | 200819_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS15 | 200819_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0054
p=0.4367
p=0.4682
p=0.1958
p=3e−04
p=0.0337
p=0.1781
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2